Region-wise Causal Analysis of Epistemic Visual Uncertainty for Object Hallucination in Large Vision-Language Models

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Abstract

Large vision-language models (LVLMs) can hallucinate objects that are not present in an image, limiting their reliability in applications that require accurate visual grounding. Prior work has shown that epistemic uncertainty in visual representations is associated with hallucination, but it remains unclear which semantic image regions causally contribute to these errors. This work introduces a region-wise causal intervention framework that suppresses epistemically uncertain visual tokens within selected semantic regions and measures the resulting change in hallucination.

A ROHE-style removed-object benchmark is constructed from 522 MS COCO images, each containing one target object that is removed using LaMa inpainting. Object-presence decisions are evaluated using POPE-style yes/no questions under five masking conditions: no masking, global masking, removed-object masking, context masking, and background masking.

The framework is implemented for LLaVA-1.5-7B, LLaVA-1.5-13B, and Qwen2.5-VL-7B. The evaluation further includes matched random controls, matched low-uncertainty controls, active-suppression ablations, original-image sanity checks, and paired bootstrap significance testing.

Global and background masking reduce hallucination in LLaVA-1.5-7B and Qwen2.5-VL-7B, whereas the same intervention does not improve LLaVA-1.5-13B. The control experiments show that the improvements are not uniformly specific to uncertainty-guided token selection, and the original-image sanity evaluations reveal a measurable trade-off between hallucination reduction and correct visual recognition. These findings suggest that global and background visual information can influence object-presence decisions, while the effectiveness and specificity of uncertainty-guided intervention remain model dependent.

1 Introduction

Large vision-language models (LVLMs) are increasingly used for image understanding tasks, but they can produce object hallucinations: they mention or confirm objects that are not actually present in an image. This reduces reliability, especially in settings where accurate visual grounding is important.

Recent work on epistemic uncertainty suggests that hallucination may be related to uncertainty in visual representations (Seo et al., 2025). This study extends this line of work by investigating a finer-grained causal question: *which semantic image regions contribute to hallucination when uncertain visual evidence is suppressed?* In particular, when an object is removed from an image, is hallucination primarily driven by uncertainty within the removed-object region, the surrounding context, or the broader scene background? A second question is whether the same regional pattern remains consistent across different LVLMs.

The objective is to determine whether uncertainty within specific semantic image regions has a measurable causal influence on object hallucination. This requires a setting in which the correct answer is known and the image region corresponding to the absent object is available. A removed-object setup is therefore used instead of a standard image-captioning benchmark.

Experimental design rationale. Standard CHAIR and POPE are useful hallucination benchmarks, but they are not sufficient for this intervention study. CHAIR evaluates object hallucination in generated captions, but it does not provide a controlled counterfactual image in which an object has been removed (Rohrbach et al., 2018). POPE provides binary object-presence questions, but it does not provide the removed-object mask, context region, or background region required for region-wise masking (Li et al., 2023). The

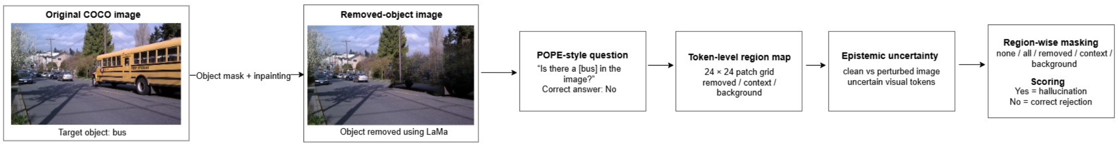

Figure 1: Overview of the region-wise causal intervention pipeline. A target object is removed from a COCO image using LaMa inpainting, and the model is asked a POPE-style object-presence question whose correct answer is known by construction. The image is mapped to a token-level region map with removed-object, context, and background regions. Epistemically uncertain visual tokens are then suppressed globally or within a selected region, and the change in hallucination rate is used as the causal effect.

evaluation therefore combines the removed-object idea from ROHE-style analysis with POPE-style yes/no questions.

A ROHE-style removed-object dataset is constructed from MS COCO (Lin et al., 2014), inspired by removed-object hallucination evaluation (He et al., 2025). For each image, one target object is removed using LaMa inpainting (Suvorov et al., 2022). The model is then asked a question such as “Is there a car in the image?” Since the queried object has been removed, the correct answer is “No”. An answer beginning with “Yes” is counted as a hallucination.

Each image is divided into three semantic regions: the removed-object region, the surrounding context region, and the remaining background. Epistemically uncertain visual tokens are suppressed either globally or within one selected region. The resulting change in hallucination rate is interpreted as the causal effect of that intervention. To test whether the effect is uncertainty-specific, uncertainty-guided masking is compared with matched random masking and matched low-uncertainty masking. The intervention is also repeated on the original, unedited images to measure its effect on correct visual recognition.

This setup enables paired causal inference under controlled interventions. If masking changes an answer from “Yes” to “No” on a removed-object image, the intervention corrects a hallucination. If the same intervention changes an answer from “Yes” to “No” on an original image, it introduces a false negative.

The main contributions are:

- A region-wise causal intervention framework for analyzing epistemically uncertain visual tokens within removed-object, context, and background regions.
- A curated ROHE-style removed-object benchmark containing 522 MS COCO samples with semantic region annotations and POPE-style object-presence questions.
- Model-specific implementations for LLaVA-1.5-7B, LLaVA-1.5-13B, and Qwen2.5-VL-7B.
- Matched random controls, matched low-uncertainty controls, active-suppression ablations, original-image sanity checks, and paired bootstrap testing.
- Evidence that global and background masking reduce hallucination in LLaVA-1.5-7B and Qwen2.5-VL-7B, but not in LLaVA-1.5-13B, and that the gains are not uniformly specific to uncertainty-guided token selection.

2 Related Work

2.1 Object Hallucination in Vision-Language Models

Object hallucination occurs when a vision-language model refers to an object that is not present in an image. CHAIR measures object hallucination in image captioning by checking whether generated object mentions are grounded in image annotations (Rohrbach et al., 2018).

POPE evaluates hallucination through binary object-presence questions (Li et al., 2023). This format provides a simple and reproducible way to classify hallucinated answers. These benchmarks motivate the evaluation style used here, but the proposed intervention requires counterfactual region annotations that are not available in standard CHAIR or POPE.

2.2 Epistemic Uncertainty and Hallucination

Prior work suggests that uncertain visual tokens are associated with object hallucination (Seo et al., 2025). If a visual representation changes substantially under perturbation, the corresponding token may reflect unstable visual grounding.

The present study asks a more specific causal question. Instead of only measuring whether uncertainty correlates with hallucination, the analysis tests whether suppressing uncertain tokens in selected semantic regions changes the model's answer.

3 Method

3.1 ROHE-style Removed-Object Dataset

A removed-object dataset is constructed from MS COCO validation images (Lin et al., 2014). Eight target categories are selected:

dog, cat, car, chair, bicycle, bus, bottle, bench

For each selected image, a suitable target-object instance is chosen, its segmentation mask is extracted, and the object is removed using LaMa inpainting (Suvorov et al., 2022). The final filtered dataset contains 522 samples.

For every removed image, a binary object-presence question is created:

Is there a [target object] in the image?

Since the target object has been removed, the correct answer is "No". A response beginning with "Yes" is counted as a hallucination.

A rule-based quality filter is used to ensure that the region-wise intervention is meaningful. Very small removed objects are excluded because they correspond to too few patch tokens for reliable region-level analysis. Very large removed objects are also excluded because they leave too little context or background. The LLaVA-oriented quality filter requires 576 total patch tokens, at least 250 background tokens, 20–150 removed-region tokens, and 30–200 context-region tokens.

3.2 Semantic Region Maps

Each image is divided into three regions:

- **Removed region**: the original target-object mask.
- **Context region**: a dilated ring around the removed object.
- **Background region**: all remaining image area.

For LLaVA, each image is mapped to a fixed 24×24 visual patch grid containing 576 patch tokens. Each visual token is assigned to the semantic region occupying the largest fraction of the corresponding image patch. This produces a token-level semantic region map aligned with the model's visual representations.

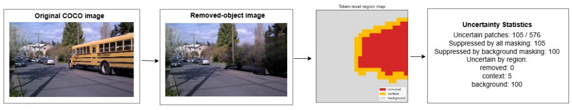

Figure 2: Example ROHE-style sample. The original image contains the target object, while the removed-object image is generated using LaMa inpainting. The removed-object mask is converted into removed-object, context, and background token regions.

For Qwen2.5-VL, semantic regions are assigned to dynamically generated visual-token grids rather than a fixed 24×24 grid. Region assignments are computed after the model-specific visual geometry has been determined.

Figure 2 illustrates one example.

3.3 Epistemic Uncertainty Framework

The method builds on the epistemic uncertainty framework of Seo et al. (2025). Uncertainty is estimated by comparing clean visual representations with adversarially perturbed representations in feature space. Tokens whose representations change substantially under perturbation are treated as epistemically uncertain.

The existing implementation is extended to support the ROHE-style dataset and region-wise causal intervention. The extensions include semantic region maps, token-to-region alignment, region-specific masking conditions, paired evaluation, and model-specific Qwen support.

3.4 LLaVA Backend

The LLaVA backend follows Seo et al. (2025). Images are encoded into a fixed 24×24 patch grid containing 576 visual tokens. Representation-space perturbations are generated using projected gradient descent, and uncertainty is estimated by comparing clean and perturbed hidden representations. Uncertain visual tokens are suppressed inside selected visual-attention layers before language generation.

3.5 Qwen2.5-VL Backend

Qwen2.5-VL uses a different visual architecture from LLaVA (Bai et al., 2025). Instead of a fixed visual-token grid, it generates a dynamic grid whose dimensions depend on the input image. Neighboring patches are merged before entering the language model, producing a variable number of merged visual tokens.

A separate Qwen backend is therefore implemented. Dynamic visual-token grids are extracted from the image processor, semantic regions are projected onto the merged-token grid, and representation-space perturbations are generated from the Qwen visual representations. Uncertainty is computed by comparing clean and perturbed merged-token representations. Rather than masking inside visual-attention layers, uncertain merged visual embeddings are suppressed immediately before language-model input.

Although the intervention point differs, the causal conditions and within-model effect calculation remain the same.

3.6 Region-wise Causal Intervention

Five conditions are evaluated:

- **None**: no masking.
- **All**: suppress all uncertain visual tokens.
- **Removed**: suppress uncertain tokens inside the removed-object region.
- **Context**: suppress uncertain tokens inside the context region.
- **Background**: suppress uncertain tokens inside the background region.

The causal effect is defined as:

$$\text{Effect} = H_{\text{none}} - H_{\text{masked}}, \quad (1)$$

where H is the hallucination rate. Positive values indicate hallucination reduction.

Three controls are included. Matched random masking suppresses the same number of randomly selected tokens from the same eligible region. Matched low-uncertainty masking suppresses the same number of lowest-uncertainty tokens. Original-image sanity evaluation repeats the intervention on images in which the queried object is present.

4 Experimental Setup

4.1 Implementation Verification

Each stage is first verified on a three-sample smoke subset. The checks cover region-map generation, visual-token alignment, representation-space perturbation, uncertainty computation, masking, and answer generation.

For the full experiments, each condition is verified to produce one output per sample. Qwen-specific checks additionally confirm that dynamic visual grids, merged-token counts, and semantic region assignments remain aligned.

4.2 Models

Three LVLMs are evaluated:

- LLaVA-1.5-7B (Liu et al., 2023)
- LLaVA-1.5-13B (Liu et al., 2023)
- Qwen2.5-VL-7B (Bai et al., 2025)

LLaVA-1.5-7B is the main development model. LLaVA-1.5-13B provides a within-family model-size comparison. Qwen2.5-VL-7B provides a cross-family comparison using a separate backend.

MiniGPT-4 and Shikra were explored during preliminary experimentation but excluded from the final quantitative evaluation because their generations were not sufficiently reliable for the controlled yes/no task.

4.3 Evaluation

All main removed-image experiments use the same 522 samples. For removed images, the correct answer is always “No”. Responses beginning with “Yes” are counted as hallucinations, and responses beginning with “No” are counted as correct rejections.

The evaluation reports hallucination rate, causal effect relative to no masking, paired bootstrap confidence intervals, answer flips, active-suppression results, matched random controls, matched low-uncertainty controls, and original-image sanity results.

4.4 Statistical Analysis

Paired bootstrap resampling is performed over the 522 image-question pairs. Each bootstrap sample preserves the pairing between the no-masking and masked outputs. Effects are reported in percentage points together with 95% confidence intervals and approximate two-sided bootstrap p-values.

5 Results

The evaluation addresses four questions: whether masking reduces hallucination, which semantic

Condition	Yes	No	Halluc. (%)	Effect (pp)
None	415	107	79.50	0.00
All	396	126	75.86	+3.64
Removed	409	113	78.35	+1.15
Context	412	110	78.93	+0.57
Background	399	123	76.44	+3.07

Table 1: LLaVA-1.5-7B hallucination results on 522 removed-object samples.

Condition	Effect (pp)	95% CI	p
All	+3.64	[1.53, 5.94]	0.0014
Removed	+1.15	[0.00, 2.30]	0.0642
Context	+0.57	[-0.57, 1.72]	0.3840
Background	+3.07	[0.96, 5.17]	0.0036

Table 2: Paired bootstrap results for LLaVA-1.5-7B.

Condition	Yes	No	Halluc. (%)	Effect (pp)
None	463	59	88.70	0.00
All	466	56	89.27	-0.57
Removed	466	56	89.27	-0.57
Context	464	58	88.89	-0.19
Background	466	56	89.27	-0.57

Table 3: LLaVA-1.5-13B hallucination results.

Condition	Effect (pp)	95% CI	p
All	-0.57	[-2.87, 1.53]	0.6588
Removed	-0.57	[-1.72, 0.57]	0.4016
Context	-0.19	[-1.15, 0.57]	0.8472
Background	-0.57	[-2.68, 1.53]	0.6800

Table 4: Paired bootstrap results for LLaVA-1.5-13B.

regions matter most, whether the effect transfers across models, and whether any improvement is specific to uncertainty-guided selection.

5.1 LLaVA-1.5-7B

Tables 1 and 2 show that global and background masking produce the strongest reductions in hallucination for LLaVA-1.5-7B. Global masking reduces hallucination by 3.64 percentage points, from 79.50% to 75.86%, with a 95% confidence interval of [1.53, 5.94]. Background masking produces a similar reduction of 3.07 percentage points, with a 95% confidence interval of [0.96, 5.17].

By contrast, removed-region masking reduces hallucination by only 1.15 percentage points, while context masking produces a reduction of 0.57 percentage points. Their confidence intervals include or touch zero, indicating that these effects are not statistically reliable under the present evaluation.

The regional pattern suggests that hallucination is influenced more strongly by distributed scene-level evidence than by uncertainty localized at the original target-object position. In particular, the background may contain co-occurring objects, scene semantics, or contextual cues that continue to support the removed object's presence.

5.2 LLaVA-1.5-13B

Tables 3 and 4 show that LLaVA-1.5-13B does not reproduce the positive masking effects observed in the 7B model. The baseline hallucination rate is 88.70%, and all masking conditions either leave the rate nearly unchanged or increase it slightly.

Global, removed-region, and background masking each produce an effect of -0.57 percentage points, while context masking produces an effect of -0.19 percentage points. All bootstrap confidence intervals include zero, and none of the corresponding p-values indicates a reliable improvement.

This result shows that the effectiveness of the intervention is not determined by model size alone. The larger model may rely on different visual representations or multimodal fusion strategies.

5.3 Qwen2.5-VL-7B

Tables 5 and 6 show that Qwen2.5-VL-7B exhibits a regional pattern similar to LLaVA-1.5-7B. Global masking reduces hallucination from 58.24% to 54.41%, corresponding to a 3.83 percentage-point reduction with a 95% confidence interval of [2.30, 5.56].

Background masking reduces hallucination by 3.07 percentage points, with a 95% confidence interval of [1.53, 4.60]. Both effects are statistically supported.

Removed-region masking produces no aggregate effect, while context masking slightly increases hallucination by 0.19 percentage points. This similarity between Qwen2.5-VL-7B and LLaVA-1.5-7B suggests that the importance of global and background visual evidence is not specific to a single visual encoder or tokenization scheme.

5.4 Control Experiments

5.4.1 Active-Suppression Ablation

The weak removed-region and context-region effects are not explained by inactive masks. For LLaVA-1.5-7B, removed and context masking remain weak even when analysis is restricted to samples with at least one suppressed token. The same

Condition	Yes	No	Unk.	Halluc. (%)	Effect
None	304	218	0	58.24	0.00
All	284	238	0	54.41	+3.83
Removed	304	218	0	58.24	0.00
Context	305	216	1	58.43	-0.19
Background	288	234	0	55.17	+3.07

Table 5: Qwen2.5-VL-7B hallucination results. Effect values are percentage points.

Condition	Effect (pp)	95% CI	p
All	+3.83	[2.30, 5.56]	< 0.0001
Removed	0.00	[-0.77, 0.77]	1.0000
Context	-0.19	[-0.96, 0.57]	0.8352
Background	+3.07	[1.53, 4.60]	< 0.0001

Table 6: Paired bootstrap results for Qwen2.5-VL-7B.

conclusion holds for Qwen2.5-VL-7B and LLaVA-1.5-13B.

5.4.2 Matched Random Control

For LLaVA-1.5-7B, five deterministic random seeds produce effects similar to high-uncertainty masking. Differences are small relative to variation across seeds. For Qwen2.5-VL-7B, the seed-42 matched random control is also statistically indistinguishable from high-uncertainty masking in every region.

These results suggest that token count, semantic location, and general information removal explain a substantial part of the observed reduction.

5.4.3 Matched Low-Uncertainty Control

The low-uncertainty control reveals different behavior across the two working backends. For LLaVA-1.5-7B, all matched low-uncertainty conditions produce the same answers as the no-masking baseline. High-uncertainty global and background masking therefore outperform matched low-uncertainty masking.

For Qwen2.5-VL-7B, high-uncertainty masking does not outperform matched low-uncertainty masking. The removed-region comparison slightly favors low-uncertainty masking. Thus, uncertainty ranking identifies influential tokens in the LLaVA-1.5-7B backend but is not validated as a selective causal signal in Qwen2.5-VL-7B.

Taken together, the control experiments show that uncertainty specificity is backend dependent. In LLaVA-1.5-7B, high-uncertainty global and background masking behaves differently from matched low-uncertainty masking, suggesting that the uncertainty ranking identifies visually influential tokens. However, the lack of a consistent advantage over matched random masking shows that uncertainty alone does not explain the entire effect.

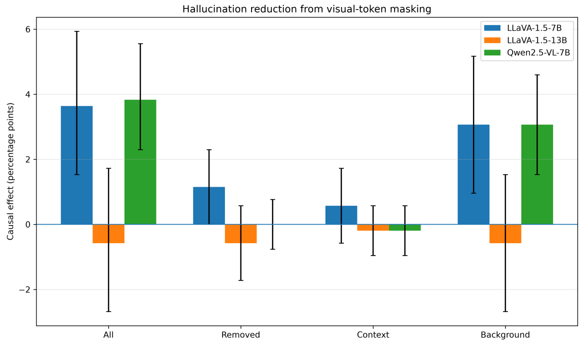

Figure 3: Hallucination reduction by model and masking condition. Positive values indicate reductions relative to each model's no-masking baseline.

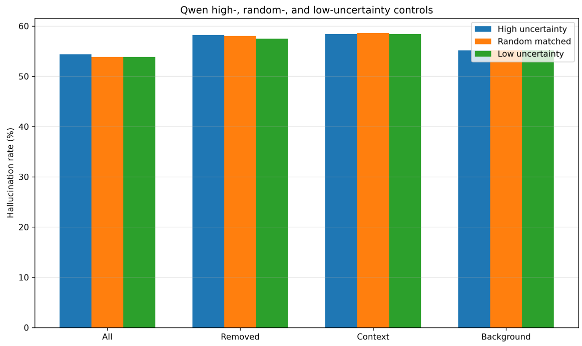

Figure 4: Qwen2.5-VL-7B high-uncertainty, matched random, and matched low-uncertainty controls.

For Qwen2.5-VL-7B, high-uncertainty masking does not outperform either matched random or matched low-uncertainty masking. The Qwen results are therefore more consistent with a general information-removal effect than with a selective uncertainty-driven intervention.

5.5 Original-Image Sanity Evaluation

For LLaVA-1.5-7B, global masking reduces accuracy by 2.30 percentage points and background masking reduces it by 1.92 points. For Qwen2.5-VL-7B, the corresponding drops are 2.68 and 1.92 points.

Table 7 and Figure 5 show that the interventions producing the largest hallucination reductions also cause statistically supported losses in correct recognition.

5.6 Cross-Model Comparison

Table 8 and Figure 3 show a similar global/background pattern in LLaVA-1.5-7B

Model	Condition	Accuracy (%)	Drop (pp)	95% CI	p
LLaVA-1.5-7B	None	94.64	0.00	–	–
LLaVA-1.5-7B	All	92.34	2.30	[0.77, 4.02]	0.0052
LLaVA-1.5-7B	Removed	94.25	0.38	[-0.38, 1.15]	0.4550
LLaVA-1.5-7B	Context	94.25	0.38	[0.00, 0.96]	0.2760
LLaVA-1.5-7B	Background	92.72	1.92	[0.38, 3.45]	0.0136
Qwen2.5-VL-7B	None	90.42	0.00	–	–
Qwen2.5-VL-7B	All	87.74	2.68	[1.15, 4.41]	0.0002
Qwen2.5-VL-7B	Removed	90.23	0.19	[-0.38, 0.77]	0.7892
Qwen2.5-VL-7B	Context	90.42	0.00	[-0.57, 0.57]	1.0000
Qwen2.5-VL-7B	Background	88.51	1.92	[0.77, 3.26]	0.0016

Table 7: Original-image sanity evaluation. Positive drops indicate reduced correct recognition after masking.

Model	Baseline (%)	All	Removed	Context	Background
LLaVA-1.5-7B	79.50	+3.64	+1.15	+0.57	+3.07
LLaVA-1.5-13B	88.70	-0.57	-0.57	-0.19	-0.57
Qwen2.5-VL-7B	58.24	+3.83	0.00	-0.19	+3.07

Table 8: Cross-model baseline hallucination rates and masking effects in percentage points.

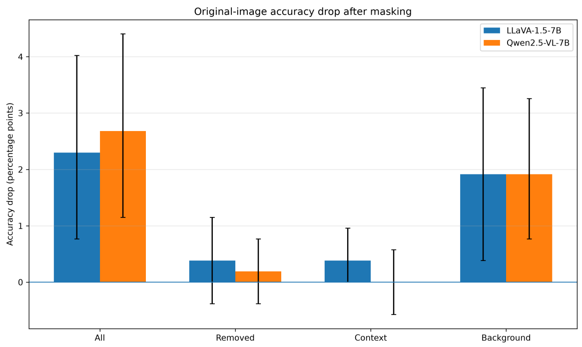

Figure 5: Original-image accuracy drop after masking.

and Qwen2.5-VL-7B. The negative LLaVA-1.5-13B result shows that this effect is not universal and remains model dependent.

5.7 Summary

Three conclusions emerge. First, global and background masking produce the strongest effects in LLaVA-1.5-7B and Qwen2.5-VL-7B. Second, LLaVA-1.5-13B does not reproduce the pattern. Third, uncertainty specificity is backend dependent: LLaVA-1.5-7B high-uncertainty masking differs from matched low-uncertainty masking but not consistently from matched random masking, while Qwen2.5-VL-7B high-uncertainty masking does not outperform either control.

6 Discussion

The results show that global and background masking, rather than removed-region or context-region masking, produce the largest hallucination reductions in LLaVA-1.5-7B and Qwen2.5-VL-7B. This pattern suggests that scene-level and co-occurrence information outside the removed-object region contributes to object-presence decisions.

Background regions may contain cues such as roads, rooms, tables, people, or other objects that are statistically associated with the queried category. Suppressing these representations can weaken misleading scene priors even when little useful evidence remains in the removed-object region itself.

The cross-model results show that the regional pattern occurs in two distinct LVLM backends, but it is not universal. LLaVA-1.5-13B produces mixed helpful and harmful answer flips, leading to no reliable net improvement.

The controls refine the interpretation. In LLaVA-1.5-7B, high-uncertainty global and background masking clearly differs from matched low-uncertainty masking, which produces no answer changes. However, high-uncertainty masking does not consistently outperform matched random masking across five seeds. In Qwen2.5-VL-7B, high-uncertainty masking does not outperform either matched random or matched low-uncertainty masking. Token count, semantic region, architecture, and general information removal therefore contribute to the effect.

These findings extend the analysis of Seo et al. (2025). Prior work showed that epistemic uncertainty in visual tokens is associated with object hallucination and that suppressing uncertain tokens can reduce hallucination. The present study adds a region-wise causal perspective and matched controls. The results show that uncertainty-guided suppression can identify influential representations in some backends, particularly LLaVA-1.5-7B, but the improvement is not uniformly specific to uncertainty. This distinction is important because correlation between uncertainty and hallucination does not necessarily imply that the highest-uncertainty tokens are uniquely responsible for the model's incorrect answer.

The original-image sanity results show that the intervention is not cost-free. The same global and background masking conditions that reduce hallucination also reduce correct recognition by approximately two percentage points. The intervention suppresses both misleading and informative visual evidence.

Overall, epistemic uncertainty remains useful as a diagnostic signal for visually unstable representations, but uncertainty-guided masking alone does not isolate uniquely causal hallucination tokens. The region-wise framework helps separate where visual evidence matters from whether uncertainty ranking is itself selective. This distinction provides a more precise basis for developing future interventions that suppress misleading visual evidence while preserving information required for correct recognition.

7 Conclusion

This study introduces a region-wise causal framework for analyzing epistemic visual uncertainty in object hallucination. The framework combines a ROHE-style removed-object benchmark, semantic region maps, representation-space perturbations, uncertainty estimation, region-specific masking, matched controls, and original-image sanity evaluation.

Global and background masking reduce hallucination in LLaVA-1.5-7B and Qwen2.5-VL-7B, while removed-region and context-region masking produce weak or negligible effects. LLaVA-1.5-13B does not reproduce the positive effect.

The controls show that uncertainty specificity is model dependent. LLaVA-1.5-7B high-uncertainty masking differs from matched low-uncertainty masking but not consistently from matched random masking. Qwen2.5-VL-7B high-uncertainty masking does not outperform either control. Original-image sanity checks additionally reveal a trade-off between hallucination reduction and correct recognition.

Region-wise causal intervention therefore provides a useful diagnostic methodology for identifying where visual evidence influences hallucination, even when the corresponding masking strategy is not universally selective. The framework offers a reproducible foundation for future causal studies of hallucination mechanisms across visual regions, uncertainty estimators, and large vision-language model architectures. Future work includes extending the framework to open-ended generation tasks, alternative uncertainty estimators, and larger families of multimodal foundation models.

Reproducibility. The implementation, analysis scripts, evaluation pipeline, and final metrics are publicly available at:

`https://github.com/PayalThangamma/region-vlm-uncertainty`

Large model checkpoints, datasets, and intermediate tensors are excluded from version control but can be regenerated using the documented pipeline.

Limitations

The evaluation uses a controlled benchmark containing 522 samples and eight object categories. The setting provides clear counterfactual labels but does not represent the full diversity of open-ended captioning or multimodal dialogue.

LaMa inpainting may introduce artifacts. However, every condition is evaluated on the same edited images, and the primary comparisons are paired.

The answer classifier uses a yes/no prefix rule. This is appropriate for the controlled task but does not capture complex hallucination behavior in free-form generation.

The evaluation covers three LVLMs and two working backend designs. Additional architectures, object categories, uncertainty estimators, thresholds, and intervention strengths are needed before broad generalization.

Ethics Statement

This work studies object hallucination in LVLMs with the goal of improving the reliability and interpretability of multimodal systems. The evaluation uses publicly available MS COCO images and automatically generated inpainted variants. No new human-subject data are collected.

The models and datasets may inherit biases from their training data. The results should therefore be interpreted as controlled diagnostic evidence rather than a complete measure of real-world safety.

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